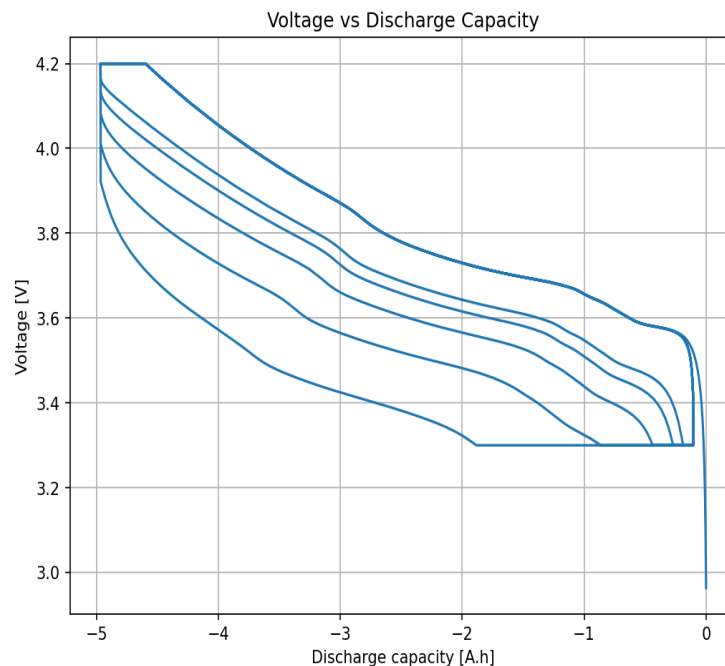


03/11/2026

Chem E 498 - Battery Design

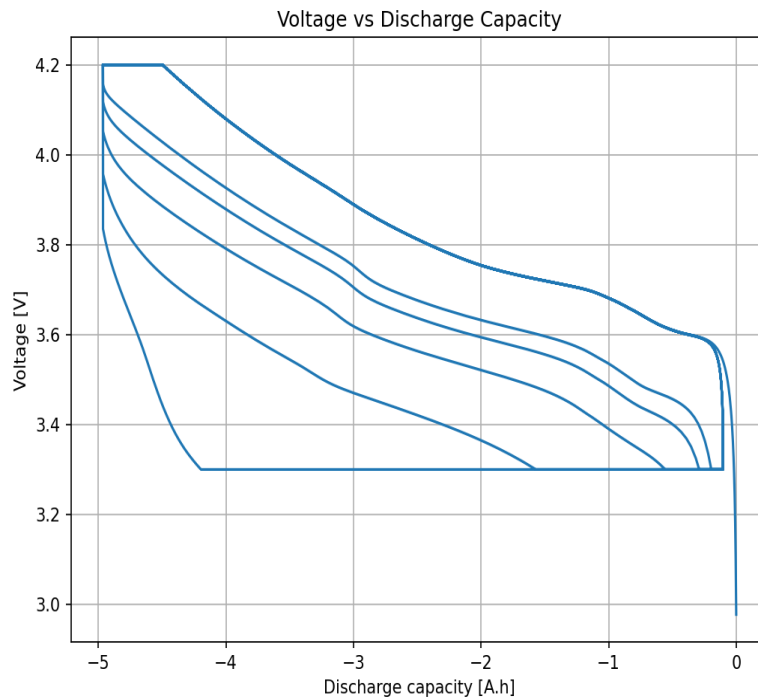
Shay Seim Tony Tran Mark Heien

1a) By expanding the list of charge and discharge cycles, run a series of simulations at C-rates of 0.5C, 1C, 2C, 4C, and 8C. Screenshot a plot of V_{cell} vs Q . Based on what you already know about various losses in a lithium battery, explain whether this cell is likely limited by separator resistance, electrode kinetics, or mass transfer inside electrode particles.



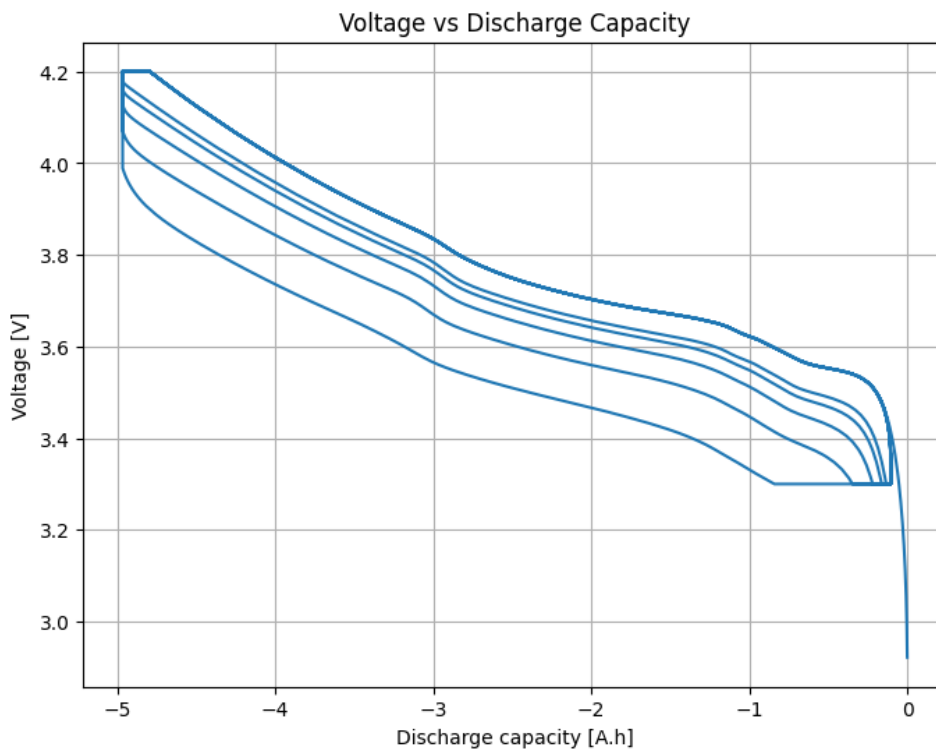
What really stands out is how the discharge curves start sloping more and more as the test goes on: the progressive droop gets worse throughout discharge which is a mass-transfer limitation inside the electrode particles.

1b) Increase the thickness of the electrodes by a factor of 2. In order to keep total cell capacity the same, also reduce the cell width by 2x (thicker, but less total cell area). Repeat the same experiment (0.5C, 1C, 2C, 4C, and 8C). Screenshot a plot of V_{cell} vs Q . Do the changes you see in (b) vs. part (a) make sense given your hypothesis in (a) about what is limiting cell performance?



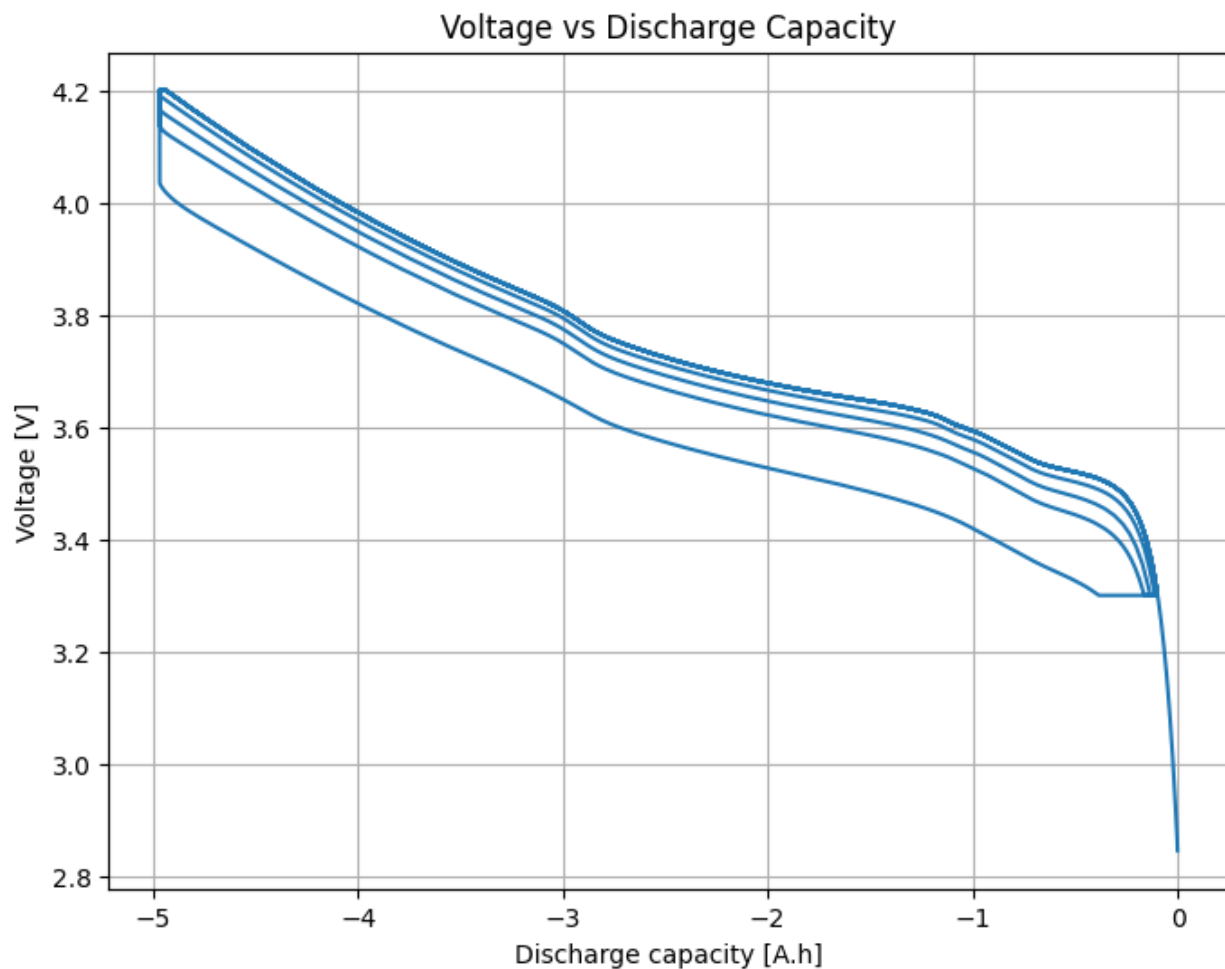
Yes, if the cell is limited by diffusion inside the individual particles, changing electrode thickness barely moves the needle as the particles don't change at all. The slight extra drop at high rates is from the increased electrode phase transport: double the thickness and it doubles the Li path length through the pore network.

1c) Returning to your original electrode thickness, reduce the size of the electrode particles by a factor of 2. Repeat the same experiment (0.5C, 1C, 2C, 4C, and 8C). Screenshot a plot of V_{cell} vs Q . Do the changes you see in (c) vs. part (a) make sense given your hypothesis in (a) about what is limiting cell performance?

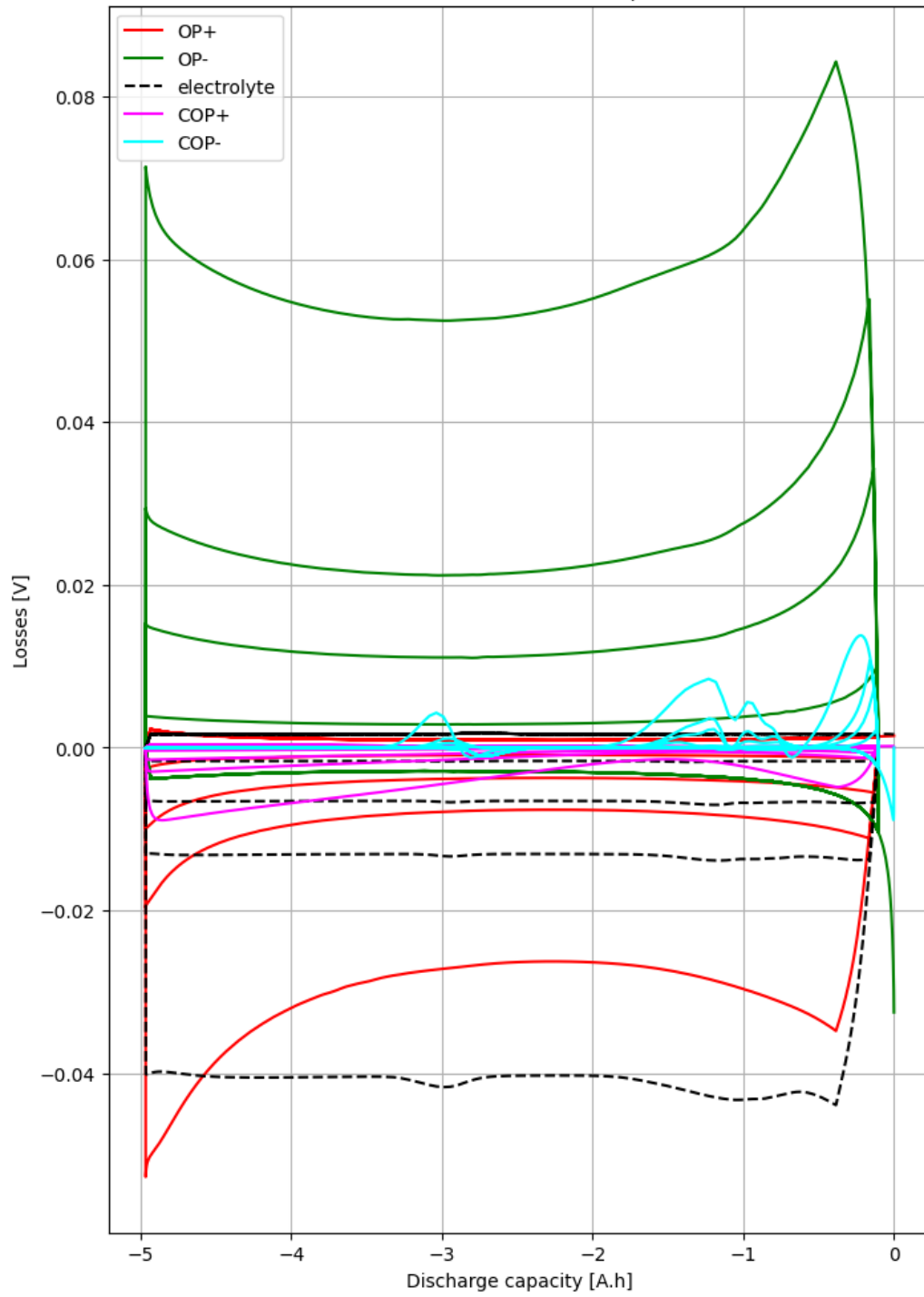


Yes, this makes sense. Halving the radius shortens the diffusion path by 2X, and diffusion time scales as r^2/D , the equilibration time drops by 4X. This nails the limiter we already flagged (COP+). Smaller particles also give a 2X higher specific surface area, which cuts the local current density and knocks down the kinetic overpotentials. The loss breakdown shows COP+ is reduced by roughly 4X vs. Part(a). This backs up the conclusion that solid-state diffusion in the + particles is the primary performance limiter.

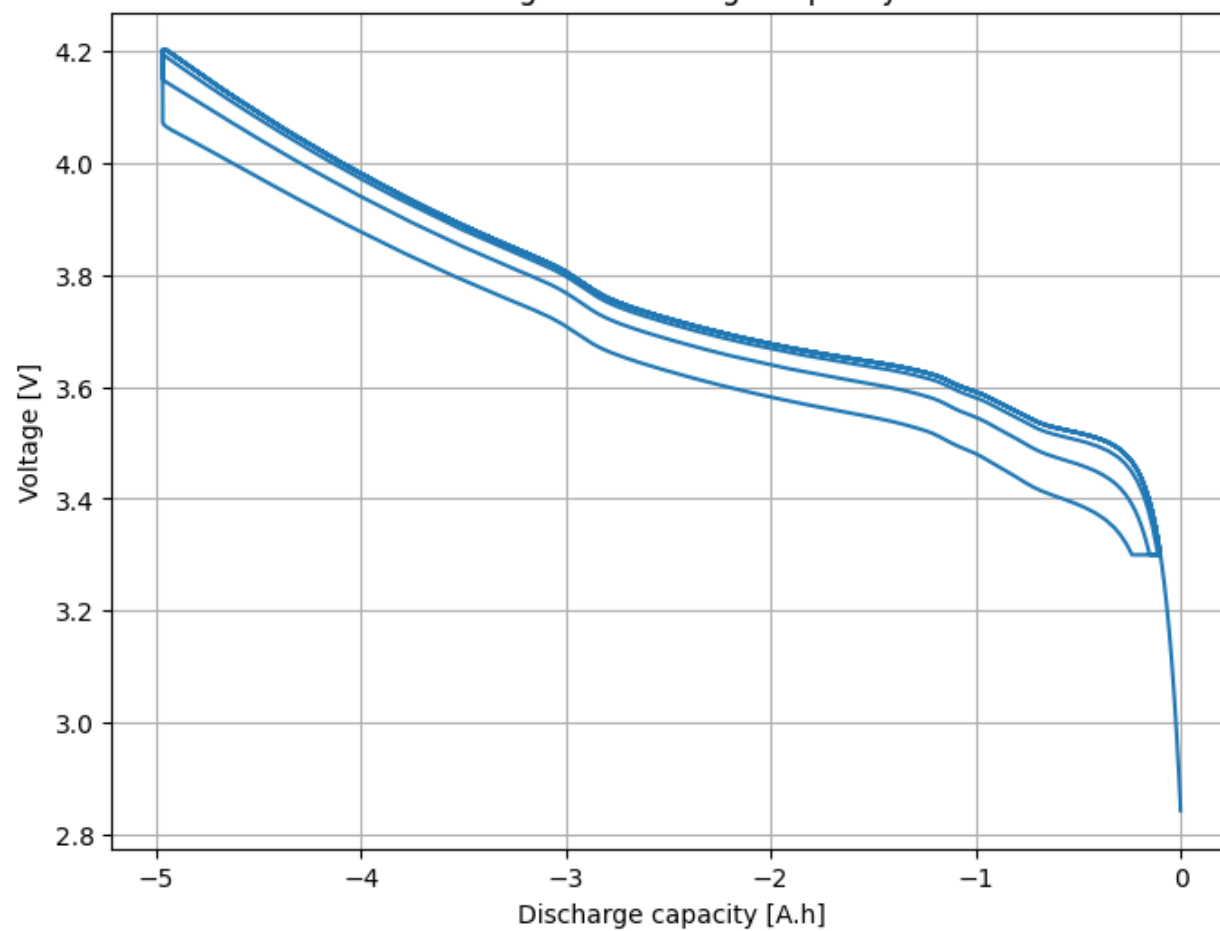
1d) By plotting various contributions to $V-V_{eqn}$, verify your conclusions in parts (a)~(c). Assuming electrode particles can be decreased in size by up to a factor of 10, and the electrodes can be printed with any thickness, explain whether the default Mohtat 2020 cell is close to optimized in terms of performance in a drone application requiring 25C discharge rates. If not, modify the cell to make a good 5 A-hr drone battery. How much total cell area is needed in such a cell?

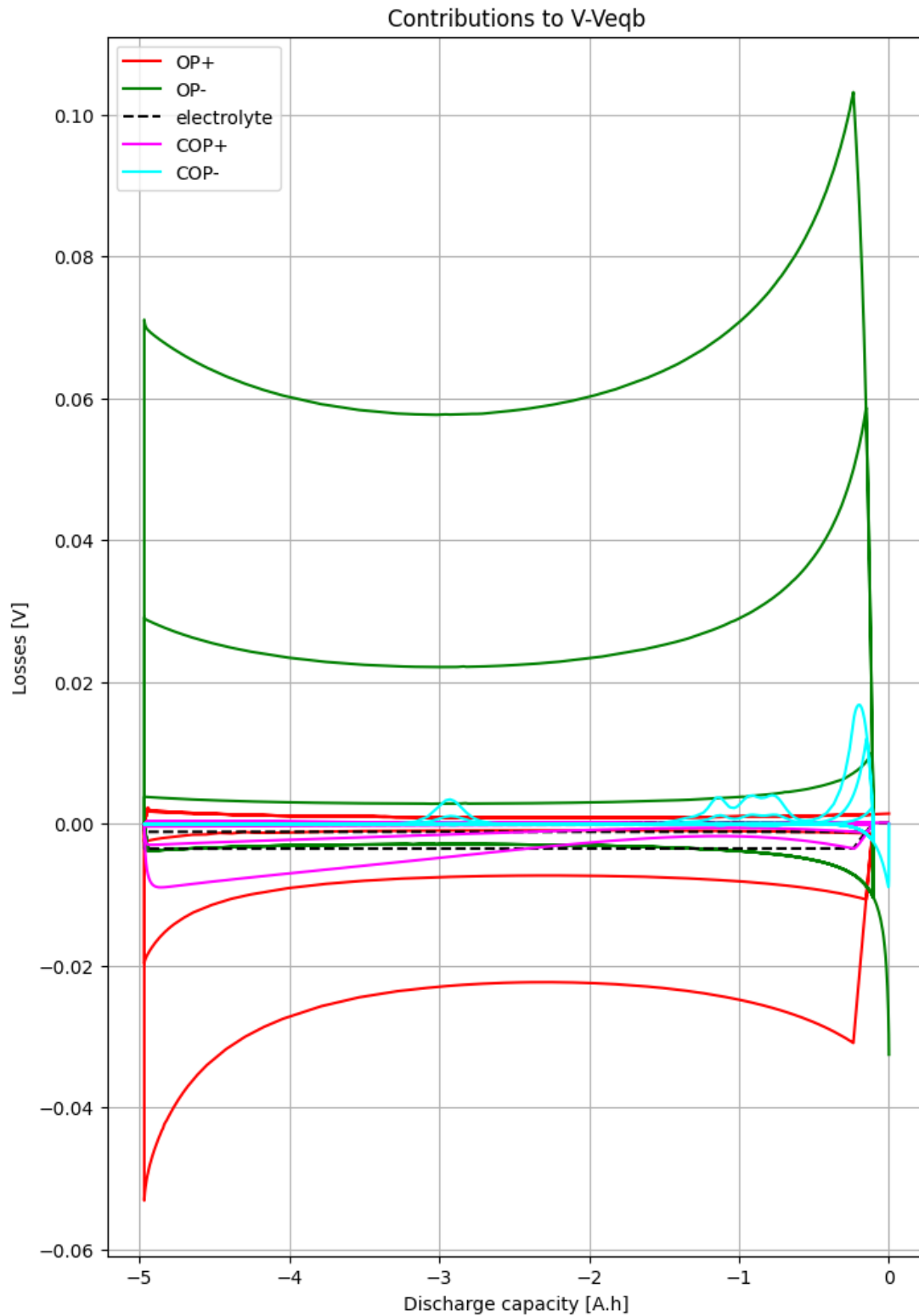


Contributions to V-Veqb



Voltage vs Discharge Capacity





With 10X smaller particles, COP+ is pretty much wiped out at 25C. The whole loss picture shifts over to kinetic overpotentials (OP-) and electrolyte ohmic losses.

The 0.5X thickness configuration (Area = $0.0205/0.5 = 0.41 \text{ m}^2$) delivers near full capacity at 25C.

Dropping to 0.1X thickness (Area = $0.0205/0.1 = 2.05 \text{ m}^2$) only gives a tiny bit more improvement, thus diminishing returns.

Recommendation for a 5 Ah drone battery:

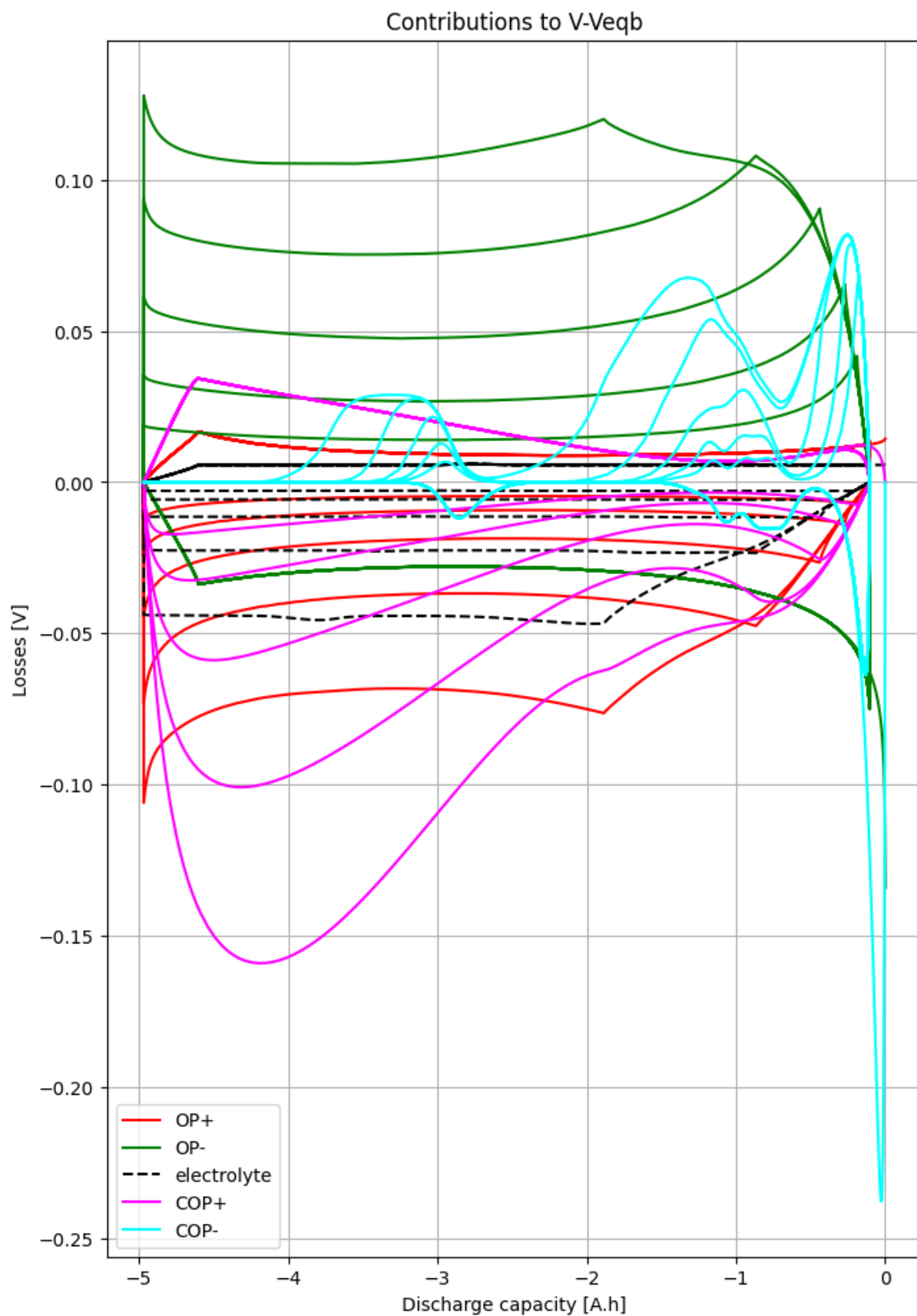
- Particle radii (r_p) = 0.35 μm , r_n = 0.25 μm (10X reduction)

- Electrode thickness: 0.5X (positive 33.5um, negative 31 um)
- Total cell area: 0.41 m² Area = 0.0205/0.5 =

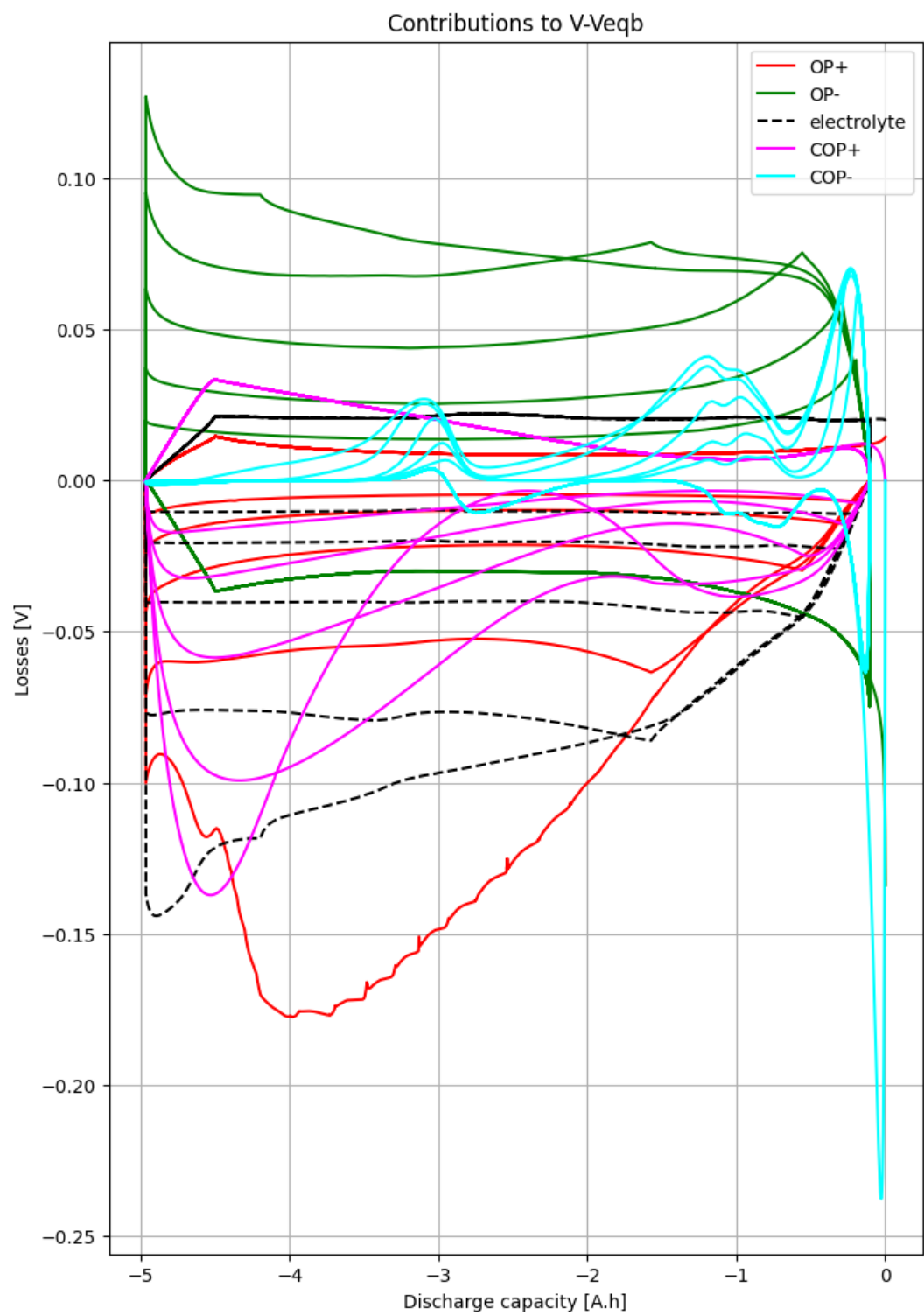
To push it further you'd need to improve electrode kinetics instead of geometric changes, since the bottleneck has now shifted from diffusion to kinetics.

Other plots generated in Python.

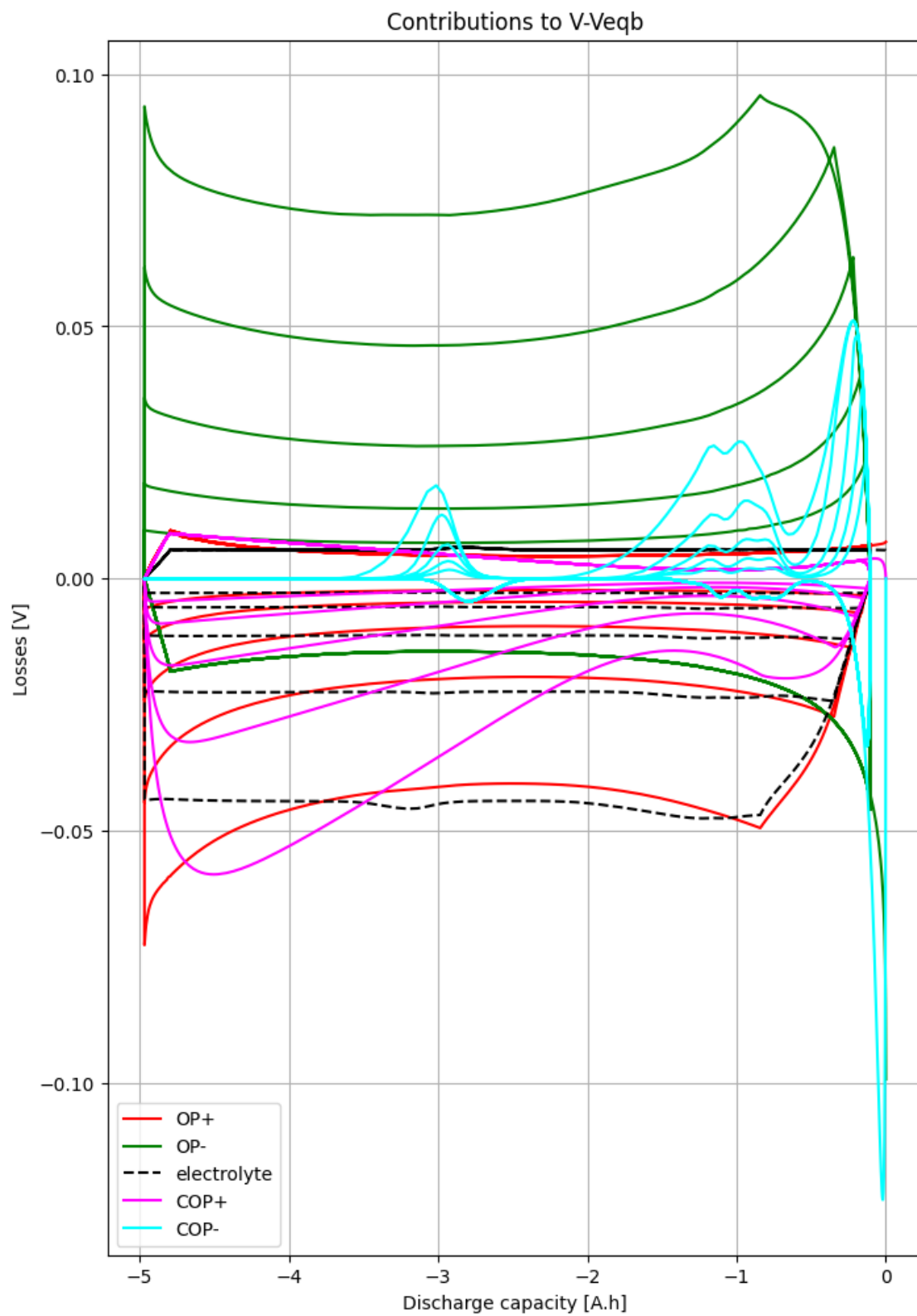
1a)



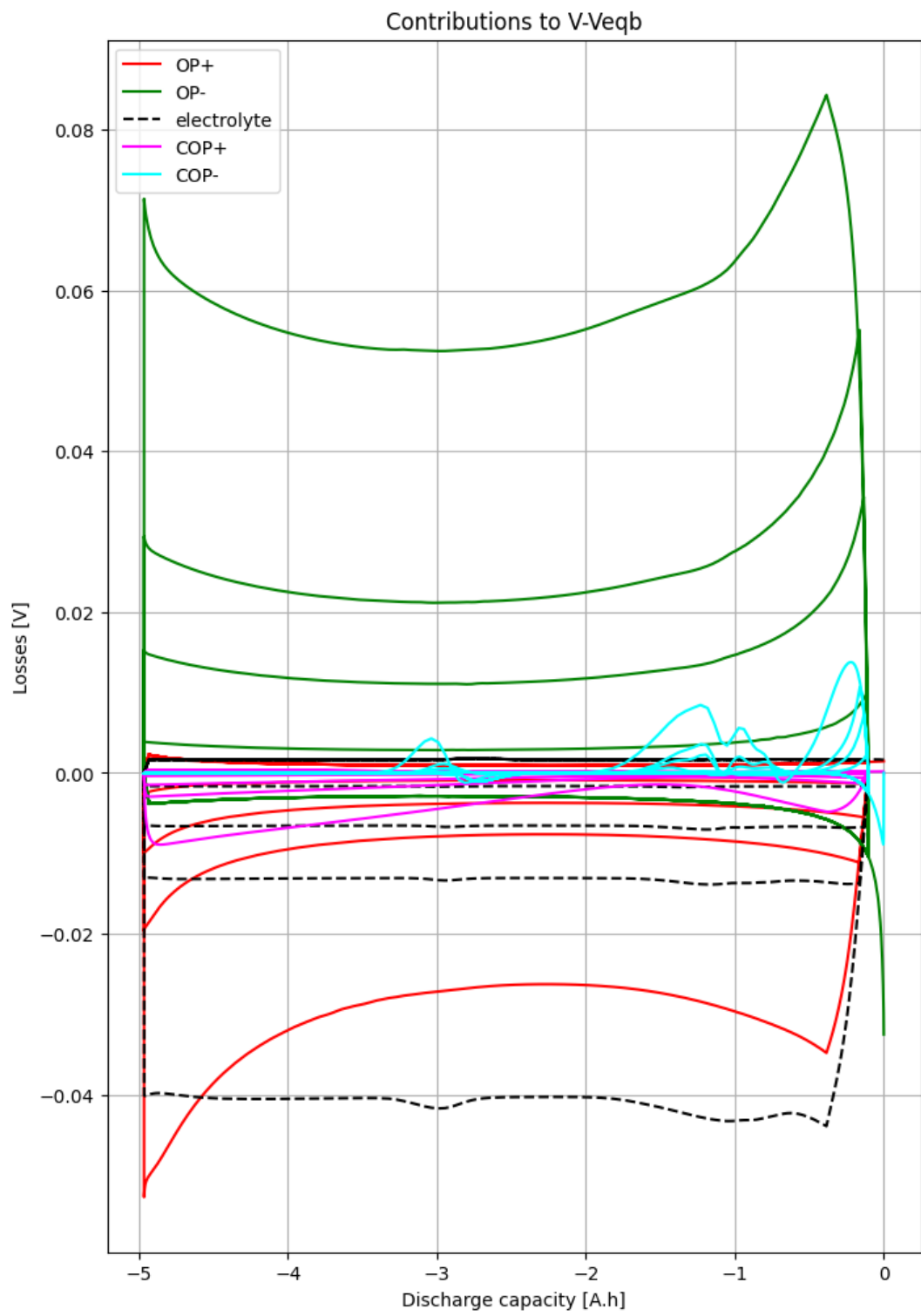
1b)



1c)



1d)



Contributions to V-Veqb

